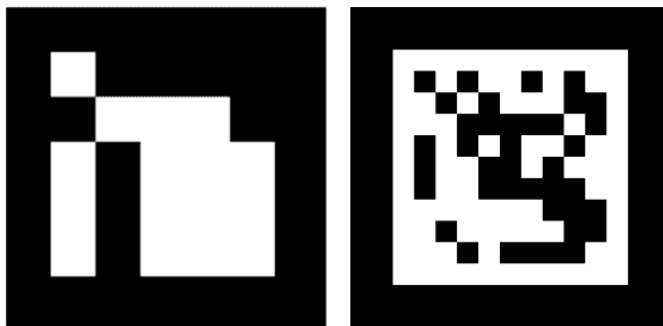


Introduction to Virtual Reality, 2026 Spring

Assignment 1-1: 360 Video

1. Rent one of the five 360 degree cameras (1 Ricoh, 1 GoPro, 2 Samsung, etc.) from the DXP Lab (they are ready to be rented), and shoot a short footage of some interesting site (maybe something that can be used as a backdrop for a blind date where virtual flirting may occur). You can refer to the Internet for the instructions as how to use the camera and make the proper file format conversion. Return the camera to the Lab according to the schedule below so that the next person/team can use it in a timely manner.

Plant few AR markers (see below for examples) in the environment so that we can augment the 360 degree video later on. You might have to print and prepare "large" sized markers so that it can later be detected easily in the video (possibly taken from some distance). So you would have to study how to choose and use a particular AR tool (compatible with Unity) and examine what kind of markers to use. You can use natural images for markers if the AR tool you plan to use later can recognize them.



2. Put the content up on Youtube, and also make it such that it can be viewed on a smart phone (if your own model somehow does not support 360 video playing, you can try it on your close friend's phone).

Assignment 1-2: Unity Practice

1. Create a simple virtual environment using Unity. First import the 360 degree video as an environment video texture mapped on a large sphere. Also place few interesting "static" objects inside the sphere and extend the "virtual blind date" (class project). Good examples might be some basic terrain or ground, and objects for decorating the place. Create a desktop application that can play/view this scene and navigate within this virtual world (just using the default keyboard interface).
2. Port the content to a head set environment (smartphone/Google cardboard, or other head sets available at the lab. Check out procedures for the headsets will be announced later. Google Cardboards are available at the DXP lab, but also they are cheap enough if you want to buy them.

360 degree video Check-out Schedule:

Each group will share the cam for the given week. For example, **김민규, Arthur and 이혁민** will check out **Cam 1** and share it during **Mar 23-29**. Thus, they have to communicate with one another (email). However, each person will shoot **one hw video for oneself** (this is not team homework).

Camera	이름/이메일	Rent Schedule
1	김민규 kmg2917@naver.com PETRAS ARTHUR arthur.petras@gmail.com 이혁민 hyeokmin.lee@wrl.onl	Mar 23 ~ Mar 29
2	장윤서 siennachang@korea.ac.kr 정경서 wanna4470@naver.com	Mar 23 ~ Mar 29
3	문이빈 helene100@naver.com	Mar 23 ~ Mar 29
4	박종찬 dmdehf123@naver.com 박현준 hj.p4rk@gmail.com	Mar 23 ~ Mar 29
1	REN SHU rich_925@naver.com 강예찬 kyccj@naver.com 강자연 jayeonkang0213@gmail.com	Mar 30 ~ Apr 5
2	고가은 jjsk3218@naver.com 김경민 kgm031189@gmail.com 김민영 kul Smy25@korea.ac.kr	Mar 30 ~ Apr 5
3	김선영 sunnykim94@gmail.com 김유진 zinzinyou@korea.ac.kr 김진원 jwkim@krm.co.kr	Mar 30 ~ Apr 5
4	김혜미 hams2401@naver.com 노태준 taejun802@korea.ac.kr 문정민 strn18@naver.com	Mar 30 ~ Apr 5
1	박승현 danbaram0420@naver.com 안상준 ics8088@korea.ac.kr 엄태상 sendmethere@naver.com	Apr 6 ~ Apr 12
2	이도현 xxrcy32@gmail.com 이동규 tony7078@naver.com 이시현 go122345@naver.com	Apr 6 ~ Apr 12
3	이창엽 ckd248@naver.com 임재혁 imhyuk98@korea.ac.kr	Apr 6 ~ Apr 12
4	홍성은 youj3333@gmail.com 황주영 05judy02@gmail.com	Apr 6 ~ Apr 12

What to submit and due date:

- Youtube URL for 360 degree video: Due one week after the camera return date (post on KULMS QA board).

Due: One week after your camera return date.

- Unity project file zipped and submitted to KULMS.
Short video demonstrating the scene on desktop and/or mobile environment

Due: April 20

(if you have trouble with the camera, you can just use any 360 degree video (e.g. from the internet) to be pasted on the spherical environment temporarily).